

Little Red Dots: Super-Eddington Wind-Shrouded Engines

Type IIIn supernovae and eta Carinae's Great Eruption offer a unified picture for JWST's enigmatic Little Red Dots.

arXiv:2606.30711 — Little Red Dots as intermediate-mass, super-Eddington engines.

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Background

JWST's Little Red Dots show blackbody-like continua, broad Balmer lines, and Balmer breaks. Similar features appear in Type IIIn supernovae and eta Carinae's Great Eruption.

Are Little Red Dots overmassive black holes or something else? Their extreme luminosities and lack of variability challenge standard AGN models.

What this paper is selling

The authors propose that dense, slow winds from a central engine create a pseudo-photosphere, mimicking a virial broad-line region. This wind-like physics may explain the entire LRD life cycle, from enshrouded youth to massive black hole seeds.

Key results to remember

- LRDs, Type IIIn SNe, and the Great Eruption share similar temperatures and line profiles.
- Escape velocity arguments constrain LRD engine masses to below 10^5 solar masses.
- Super-Eddington engines with intermediate masses can reproduce LRD luminosities and spectra.

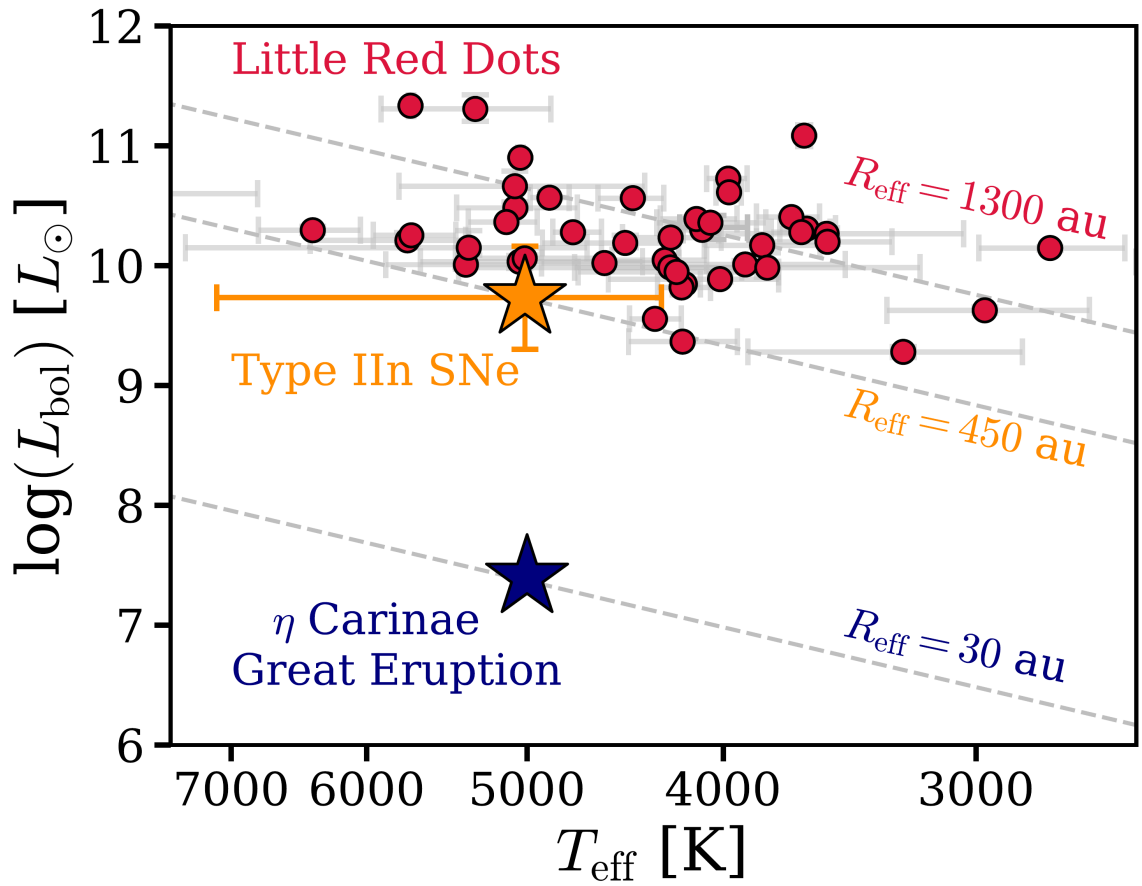


Figure 4 | LRDs, Type IIIn SNe, and the GE span 4 dex in bolometric luminosity (L_{bol}) but cluster in effective temperature (T_{eff}).

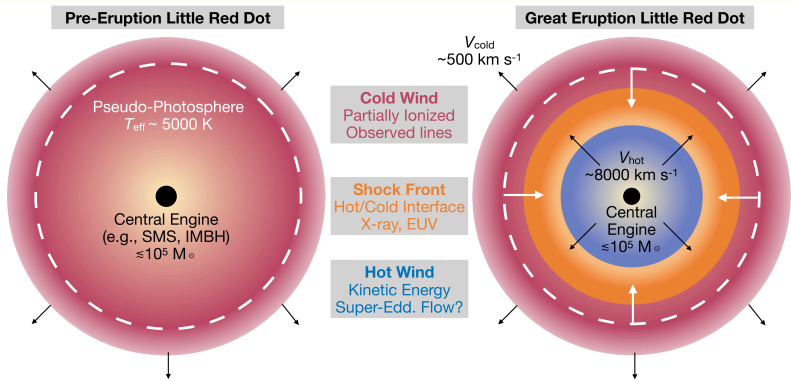


Figure 8 | Schematic of the proposed LRD scenario inspired by Type IIIn SNe and the GE.